Lecture #8: Second Derivative, Differentiable Functions & Limits Revisited

MATH 145 – Calculus I

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The Second Derivative

The derivative of f, f', is just another function, related to f.

What if we took the derivative of the derivative function?

We get what is called the **second derivative**.

($f'(x)$ is called the **first derivative**.)

The Second Derivative

The **second derivative, $f''(x)$**, is defined by the limit definition of the derivative of the first derivative. That is,

$$f''(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$$

$f''(x)$ is sometimes written as $\dfrac{d^2}{dx^2} f(x)$.

The n^{th} Derivative

We can keep this process going, taking the derivative of the second derivative to get the **third derivative**, $f'''(x)$, take the derivative of that to get the **fourth derivative**, $f^{(4)}(x)$, and so on and so on.

The n^{th} **derivative of** f is denoted as $f^{(n)}(x)$.

Example: The Second Derivative

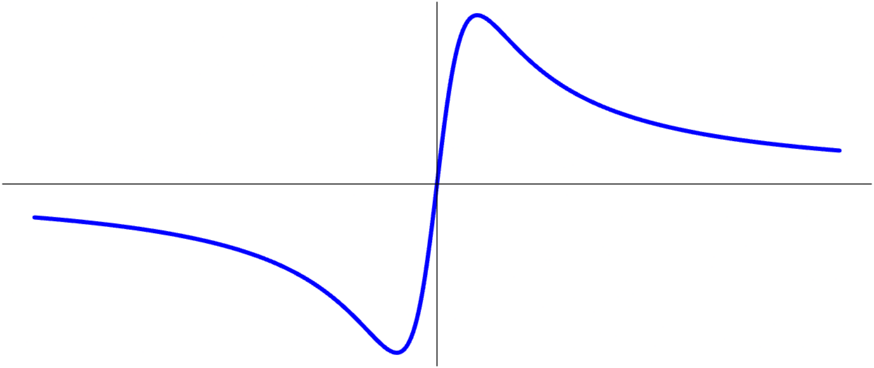
Example: Find the second derivative of $f(x) = 4x - 9$.

Example: Find the second derivative of $f(x) = 3x^2 + 5x + 7$.

Example: Let $g(x) = \frac{1}{x}$. Find $g''(4)$.

The Second Derivative, Graphically

Example: Sketch the graph of $q''(x)$ if $q(x)$ has the graph below.



Concavity of a Function

There is more to describing the shape of a graph other than just whether it is increasing or decreasing.

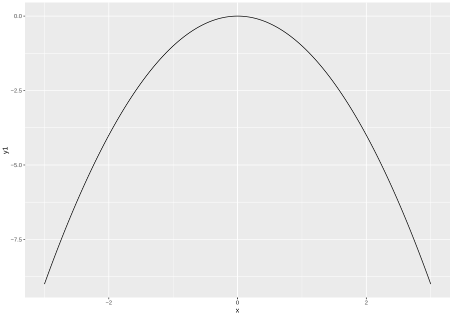


Figure 1: $a < 0$

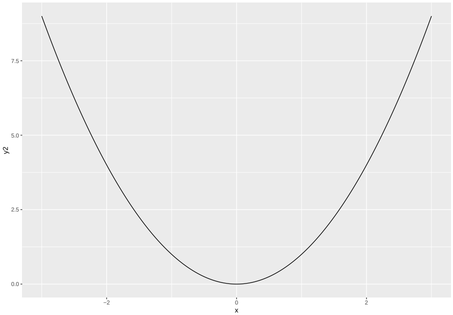


Figure 2: $a > 0$

Concavity of a Function

Let f be a differentiable function on an interval (a, b). Then f is **concave up on** (a, b) if and only if f' is increasing on (a, b).

Let f be a differentiable function on an interval (a, b). Then f is **concave down on** (a, b) if and only if f' is decreasing on (a, b).

Concavity of a Function

Recall that:

Let f be a function that is differentiable on an interval (a, b). If $f'(x) > 0$ for every x such that $a < x < b$, then f is **increasing on (a, b)**.

Let f be a function that is differentiable on an interval (a, b). If $f'(x) < 0$ for every x such that $a < x < b$, then f is **decreasing on (a, b)**.

Concavity of a Function

Combining these definition gives us that:

- $f''(x)$ is positive on (a, b) if and only if $f(x)$ is concave up on (a, b).
- $f''(x)$ is negative on (a, b) if and only if $f(x)$ is concave down on (a, b).

Relative Extrema and Inflection Points

The point where a graph f switches from increasing to decreasing or vice versa is called a **relative extreme**.

If f goes from increasing to decreasing, the relative extreme is called a **maximum**.

If f goes from decreasing to increasing, the relative extreme is called a **minimum**.

An **inflection point** is a point of a graph f where concavity changes.

Recall the Limit of a Function at a Point

Given a function f, a fixed input $x = a$, and a real number L, we say that **f has limit L as x approaches a**, and write

$$\lim_{x \to a} f(x) = L$$

provided that we can make $f(x)$ as close to L as we like by taking x sufficiently close (but not equal) to a. If we cannot make $f(x)$ as close to a single value as we would like as x approaches a, then we say that **f does not have a limit as x approaches a**.

Recall Left-Hand Limits

We say that f **has limit** L_1 **as** x **approached** a **from the left** and write

$$\lim_{x \to a^-} f(x) = L_1$$

provided that we can make the value of $f(x)$ as close to L_1 as we like by taking x sufficiently close to a, while always having $x < a$. We call L_1 the **left-hand limit of** f **as** x **approaches** a.

Recall Right-Hand Limits

We say that f **has limit L_2 as x approached a from the right** and write

$$\lim_{x \to a^+} f(x) = L_2$$

provided that we can make the value of $f(x)$ as close to L_2 as we like by taking x sufficiently close to a, while always having $x > a$. We call L_1 the **right-hand limit of f as x approaches a**.

Recall Left-Hand and Right-Hand Limits

A function f has limit L as $x \to a$ if and only if

$$\lim_{x \to a^-} f(x) = L = \lim_{x \to a^+} f(x)$$

Non-Differentiable Functions

Example: Show that the derivative of $f(x) = |x|$ does not exist at $x = 0$.

The derivative of $f(x)$ will not exists at a point $x = a$ if there is a "cusp", or corner or sharp turn, on the graph there.

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- $\lim\limits_{x \to a} f(x) = f(a)$.

Differentiability

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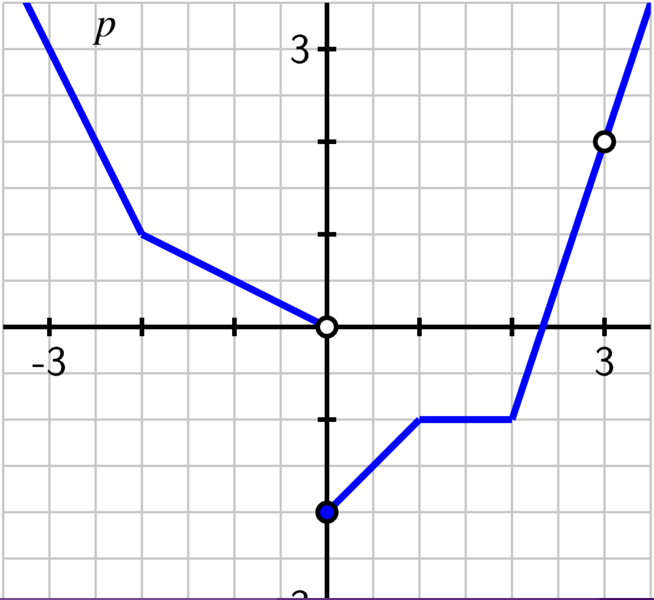
- If f is differentiable at $x = a$, the f is *locally linear* at $x = a$. That is, when a function is differentiable, it looks linear when viewed up close because it resembles its tangent line there.

Example: Continuity and Differentiability



Local Linearization

If $f(x)$ is differentiable at $x = a$, the **local linearization** (or **tangent line approximation**) to $f(x)$ near $x = a$ is the function

$$y = L(x) \approx f(a) + f'(a)(x - a)$$

If $f(x)$ is differentiable at $x = a$, you can approximate it by its tangent line.

Example: Local Linearization

Example: Let $f(x) = x^2$.

Find $f(2.3)$, and then estimate it with the local linearization of f using $x = 2$.